

# ECH Test Coverage Summary

## 1. Primary ECH Tests

*(Based on `mod-openssl-ech.t`)*

### 1.1 Test Setup Validation

- Self-signed TLS certificate generation verified
- Server-side ECH key and configuration fixture generation verified
- Mismatched client ECH configuration generated for rejection testing
- Malformed ECH key material generated for negative testing

### 1.2 ECH-Enabled Server Behaviour

- Lighttpd starts successfully with ECH enabled
- Server serves standard HTTPS successfully with ECH enabled

### 1.3 Non-ECH Client Behaviour

- Non-ECH client completes TLS handshake successfully
- Connection remains standard TLS (no ECH acceptance)
- HTTP 200 response returned

### 1.4 Valid ECH Behaviour

- ECH-capable client completes TLS handshake
- Valid ECH is accepted by the server
- HTTP 200 response returned for ECH request

### 1.5 Invalid / Mismatched ECH Handling

- Mismatched ECH input is rejected correctly
- Rejection classified as ECH\_REJECTED
- No server hang or crash occurs
- Server remains healthy after invalid attempt

### 1.6 Server Lifecycle (ECH Enabled)

- Server shuts down cleanly after execution

## 1.7 TLS-Only Configuration (ECH Disabled)

- Lighttpd starts successfully without ECH
- Standard HTTPS functionality confirmed

## 1.8 ECH Attempt Against TLS-Only Server

- ECH-capable client is rejected
- Response classified as ECH\_REJECTED
- Server remains stable after attempt
- Clean shutdown confirmed

## 1.9 Missing ECH Key Directory

- Server starts despite missing key directory
- ECH clients are rejected without hanging
- Rejection classified as ECH\_REJECTED
- Error logged to stderr
- Server remains stable
- Clean shutdown confirmed

## 1.10 Malformed ECH Key Material

- Server starts with malformed key file
- ECH clients are rejected without hanging
- Rejection classified as ECH\_REJECTED
- Parse error logged
- Server remains stable
- Clean shutdown confirmed

# 2. Fallback and Routing Tests

*(Based on mod-openssl-ech-extra.t)*

## 2.1 Test Setup

- Self-signed TLS certificate generation verified
- Server-side ECH configuration generated
- Malformed client ECH configuration created
- Public and hidden virtual host content configured

## 2.2 Multi-Vhost Server Behaviour

- Lighttpd starts successfully with multi-vhost ECH configuration
- Public vhost serves expected baseline content

## 2.3 Non-ECH Client Behaviour (Hidden Host)

- TLS handshake completes successfully
- Connection remains standard TLS (no ECH)
- HTTP 200 response returned
- Request falls back to public vhost
- Hidden content is not exposed

## 2.4 Valid ECH Routing Behaviour

- ECH client completes TLS handshake
- ECH is accepted when valid configuration is supplied
- HTTP 200 response returned
- Request correctly routed to hidden vhost content

## 2.5 Invalid ECH Handling

- Malformed ECH input is rejected without hanging
- No successful HTTP 200 response returned
- Explicit failure trace generated
- Server remains stable

## 2.6 Handshake State Verification

- Distinct handshake states observed for:
  - Standard TLS
  - Accepted ECH

## 2.7 Sequential Connection Stability

- Multiple sequential ECH connections (1–5) succeed
- No degradation in performance or stability observed

## 2.8 HTTP/1.1 Over ECH

- TLS handshake completes successfully
- ECH remains active during request
- HTTP 200 response returned
- Request correctly routed to hidden content

## 2.9 Server Lifecycle (Multi-Vhost)

- Server shuts down cleanly after execution

# 3. Summary

The test suite validates:

- Correct ECH acceptance, rejection, and fallback behaviour
- Robust handling of invalid configurations and malformed inputs
- Preservation of standard HTTPS compatibility

```
seanl@SensDell:~/final-year-project/lighttpd/tests$ ./mod-openssl-ech.t
1..34
ok 1 - generated TLS certificate fixture
ok 2 - generated server ECH fixture
ok 3 - generated mismatched client ECH fixture
ok 4 - generated malformed ECH fixture
ok 5 - Starting lighttpd with mod-openssl-ech.conf
ok 6 - ECH-enabled server serves HTTPS
ok 7 - non-ECH client completed TLS handshake
ok 8 - non-ECH client stayed on standard TLS
ok 9 - non-ECH client received HTTP 200
ok 10 - ECH client completed TLS handshake
ok 11 - valid ECH was accepted
ok 12 - ECH client received HTTP 200
ok 13 - mismatched ECH input was rejected without hanging
ok 14 - mismatched ECH input produced ECH rejection
ok 15 - server stayed healthy after mismatched ECH input
ok 16 - Stopping lighttpd
ok 17 - Starting lighttpd with mod-openssl-tls.conf
ok 18 - non-ECH server serves HTTPS
ok 19 - ECH-capable client was rejected when ECH was disabled
ok 20 - ECH-disabled server returned ECH rejection
ok 21 - non-ECH server stayed healthy after ECH attempt
ok 22 - Stopping lighttpd
ok 23 - starting lighttpd with missing ECH keydir succeeds
ok 24 - missing ECH keydir rejected client without hanging
ok 25 - missing ECH keydir produced ECH rejection
ok 26 - missing ECH keydir was logged
ok 27 - server stayed healthy with missing ECH keydir
ok 28 - Stopping lighttpd
ok 29 - starting lighttpd with malformed ECH key file succeeds
ok 30 - malformed ECH key file rejected client without hanging
ok 31 - malformed ECH key file produced ECH rejection
ok 32 - malformed ECH key file was logged
ok 33 - server stayed healthy with malformed ECH key file
ok 34 - Stopping lighttpd
```

```
seanl@SensDell:~/final-year-project/lighttpd/tests$ ./mod-openssl-ech-extra.t
1..30
ok 1 - generated TLS certificate fixture
ok 2 - generated server ECH fixture
ok 3 - generated malformed client ECH config fixture
ok 4 - wrote distinct public and hidden vhost content fixtures
ok 5 - Starting lighttpd with mod-openssl-ech-extra.conf
ok 6 - public vhost baseline served expected content without ECH
ok 7 - ECH fallback: non-ECH client completed TLS handshake
ok 8 - ECH fallback: non-ECH handshake stayed on standard TLS
ok 9 - ECH fallback: non-ECH request returned HTTP 200
ok 10 - ECH fallback: hidden host without ECH served public vhost content
ok 11 - ECH-only host: non-ECH request did not expose hidden content
ok 12 - ECH-only host: ECH client completed TLS handshake
ok 13 - ECH-only host: ECH was accepted
ok 14 - ECH-only host: ECH request returned HTTP 200
ok 15 - ECH-only host: ECH request served hidden vhost content
ok 16 - invalid ECH config was rejected without hanging
ok 17 - invalid ECH config did not complete HTTP 200
ok 18 - invalid ECH config produced a failure trace
ok 19 - server stayed healthy after invalid ECH input
ok 20 - TLS vs ECH comparison: handshake traces differ as expected
ok 21 - sequential ECH connection \#1 stayed consistent
ok 22 - sequential ECH connection \#2 stayed consistent
ok 23 - sequential ECH connection \#3 stayed consistent
ok 24 - sequential ECH connection \#4 stayed consistent
ok 25 - sequential ECH connection \#5 stayed consistent
ok 26 - HTTP/1.1 ECH request completed TLS handshake
ok 27 - HTTP/1.1 ECH request kept ECH enabled
ok 28 - HTTP/1.1 ECH request returned HTTP 200
ok 29 - HTTP/1.1 ECH request served hidden vhost content
ok 30 - Stopping lighttpd
```